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Hypergraph Kolmogorov–Arnold Networks for station level meteorological forecasting

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ABSTRACT

Weather forecasting plays a vital role in various domains, including disaster prevention, resource management, and energy optimization. Meteorological observation data are crucial for accurate weather forecasting, as they provide the spatial and temporal information needed to capture complex weather patterns. Graph Neural Networks (GNNs), due to their ability to handle non-Euclidean spatial relationships, can model the spatial dependencies among weather stations and have achieved good results in various weather prediction tasks. However, traditional graph neural networks, while effective at modeling spatial relationships between weather stations, often fall short in capturing the complex, multi-dimensional dependencies across different scales. This paper introduces a novel multi-information spatio-temporal hypergraph learning framework to overcome these limitations. By integrating neighborhood and semantic hypergraph convolutional networks, the framework effectively aggregates information from both spatially adjacent and semantically similar weather data, enabling it to capture intricate spatial and temporal features. Additionally, the Kolmogorov–Arnold Network (KAN) is introduced to enhance the model's ability to learn dynamic, high-dimensional feature representations through the use of learnable univariate functions instead of fixed linear weights. Experiments on benchmark weather datasets show that the proposed method surpasses traditional spatio-temporal graph neural networks, providing higher accuracy and robustness in predicting complex meteorological phenomena.

1. Introduction

Since ancient times, weather changes have profoundly affected human production and daily life. Whether in agricultural production, transportation, tourism, and energy management, weather conditions are key influencing factors. Consequently, the development of reliable and efficient weather forecasting systems holds immense economic, scientific, and social significance. Accurate weather forecasting not only helps people plan their activities and production rationally but also effectively improves the society's ability to cope with natural disasters, which makes the task of weather forecasting a research direction of great interest.

Meteorological factors like temperature, humidity, wind, and cloud cover offer valuable historical data and robust support for researchers analyzing weather patterns. During the past decades, numerical weather prediction (NWP) has been the dominant approach, employing numerical integration to simulate meteorological dynamics in the atmosphere and on the Earth's surface [1,2]. However, due to uncertainties in the initial conditions of differential equations, NWP predictions may not be accurate enough, especially in complex atmospheric processes [1]. In addition, NWP is also computationally demanding, whereas data-driven models can infer and generate predictions several orders of magnitude faster than traditional NWP models [3], thus enabling the generation

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of very large integrated predictions [4]. Therefore, deep learning (DL)-based weather forecasting methods have received increasing attention, with the rapid development of deep learning techniques.

Recent advances in deep learning-based weather forecasting provide an alternative to NWP-based forecasting methods and have been shown to provide higher accuracy and efficiency for nonprobabilistic forecasts. Methods based on Convolutional Neural Networks [5,6], Recurrent Neural Networks [7,8], Transformer [9,10], Diffusion Models [8,11] and Graph Neural Networks (GNNs) [12–14] are presented for medium-term weather forecasting. Among these, GNNs have attracted significant attention due to their ability to model non-Euclidean spatial relationships, making them particularly suitable for processing irregularly distributed meteorological observation data. GNNs have been applied to tasks such as precipitation forecasting, temperature prediction, and the detection of extreme weather events. By leveraging graph-based structures, GNNs can effectively capture spatial dependencies among weather stations, providing a promising approach for handling complex spatio-temporal correlations in weather data. On this basis, large-scale models such as GraphCast [15], NeuralGCM [16] and Pangu-Weather [17] have made significant breakthroughs in global medium- and long-term forecasts. These models use the high-resolution, multivariate reanalysis dataset ERA5 [18] to simulate the evolution of mesoscale and global atmospheric systems. However, they rely heavily on massive data and computing resources, limiting their applicability in data-constrained or real-time business scenarios. In contrast, our research focuses on short-term forecasting tasks (within 12 h) using station-level observational data with limited variable types and short time spans. This leads to significant differences in model objectives, data conditions, and design requirements, highlighting the need for more lightweight, local prediction models.

Despite the strengths of these approaches, several challenges persist. Modeling the spatial correlations in meteorological data remains difficult due to the advection effect, wherein meteorological attributes propagate across regions through aggregation and diffusion, creating high-order dependencies across multiple directions and nodes. The traditional graph structure, typically built on pairwise relationships, struggles to represent high-order and complex spatial correlations effectively. Meteorological data are also subject to negative noise factors, such as extreme weather events (e.g., hurricanes or extreme temperatures) that disrupt spatial dependencies, and noise generated by aging equipment or transmission delays during data acquisition. Such factors exacerbate data quality fluctuations, reducing the capability of models to extract meaningful features. Moreover, most existing methods rely heavily on geographic proximity to model spatial dependencies, which fails to account for the complexity of atmospheric motion, where distant weather stations may exhibit similar patterns. This exposes the limitations of traditional approaches in capturing the inherent spatial semantic correlations within weather data. Hypergraph Neural Networks (HGNNs) [19] offer a promising solution, as they can model higher-order relationships more effectively than other methods.

Although particularly effective at capturing higher-order dependencies, HGNNs inevitably incur substantially increased parameter complexity and computational overhead. To alleviate these computational burdens and improve overall model efficiency, Kolmogorov–Arnold Networks (KANs) [20] are employed, grounded in the Kolmogorov–Arnold representation theorem, which decomposes complex multivariate functions into compositions of one-dimensional learnable spline functions. By replacing the fixed linear weights and activation functions of traditional Graph Convolutional Networks (GCNs) with adaptive spline-based univariate functions, KANs flexibly adjust feature propagation paths to capture localized nonlinear patterns – such as abrupt pressure gradients or humidity saturation – while maintaining computational efficiency. Unlike Multilayer Perceptrons (MLPs) that rely on fixed activation functions (e.g., ReLU) and deep architectures to approximate complex mappings, KANs achieve comparable expressiveness with shallower structures by optimizing spline-based activations. This design reduces computational complexity by avoiding the quadratic parameter scaling inherent in MLPs and mitigates overfitting on high-dimensional meteorological data. Recent studies have demonstrated the superior performance of KANs in time-series forecasting tasks [21,22].

To address the challenges of modeling complex spatial dependencies, mitigating data quality fluctuations, and capturing spatial semantic correlations in meteorological data, we propose Hypergraph Kolmogorov–Arnold Networks (HyperGKAN), a sequence-to-sequence framework for station-level weather forecasting. HyperGKAN represents meteorological stations as nodes connected via both neighborhood and semantic hyperedges, enabling simultaneous modeling of local proximity and latent similarity. Within HyperGKAN, hypergraph convolutions aggregate multi-source spatial information, while integrated Kolmogorov–Arnold Networks replace fixed GCN weights and activations with learnable one-dimensional spline functions. This hybrid design reduces computational complexity by avoiding the quadratic parameter scaling of deep MLPs and offers dynamic, data-driven feature propagation, thereby overcoming the scalability limitations of conventional GCNs and hypergraph schemes and enhancing adaptability to large, complex meteorological datasets. The main contributions of this paper are summarized as follows:

- The spatial dependencies between meteorological data are modeled using hypergraphs, and the hypergraph representation can more effectively model higher-order aggregation of meteorological phenomena.
- A new multi-information spatio-temporal hypergraph learning framework is proposed, which consists of neighborhood hypergraphs and semantic hypergraph convolutional networks to capture spatial features by aggregating neighboring and similar meteorological data to adapt to the complexity of atmospheric systems.
- Kolmogorov–Arnold network is introduced to enable more dynamic feature propagation in the graph structure by exploiting the unique features of KAN, in particular the use of learnable univariate functions rather than fixed linear weights.

2. Related work

This section provides a brief overview of some of the techniques relevant to this research, mainly related to deep learning-based weather forecasting and hypergraph neural networks.

2.1. Deep learning-based weather forecasting

Deep learning-based weather forecasting models are increasingly utilized for their ability to process vast meteorological datasets and uncover patterns that enable precise and timely predictions. Compared to traditional numerical weather prediction (NWP) models [23], which rely on complex physical equations and are computationally intensive, AI-based approaches [24] demonstrate notable advantages in both predictive accuracy and efficiency. Albu et al. [25] proposed NeXtNow, a convolutional encoder–decoder architecture designed to capture both spatial and temporal features in weather data, showcasing advancements in precipitation forecasting. Karevan and Suykens [26] investigated Long Short-Term Memory (LSTM) networks for time-series forecasting, highlighting their strength in modeling temporal dependencies, but their limited ability to model spatial correlation. To further improve the modeling of spatial dependencies, Pathak et al. [9] introduced a visual Transformer [27] that integrates the Fourier neural operator with self-attention mechanisms. This approach enables the efficient processing of high-resolution weather data and captures multi-scale patterns through the use of fast Fourier transforms (FFT). Additionally, to address the specific challenges posed by spherical data, a spherical Fourier transform [28] was introduced to replace planar transforms, ensuring rotational invariance and enhancing the stability of long-term forecasts, particularly those extending beyond one year.

However, a core challenge in spatial modeling stems from the non-uniform and irregular structure of meteorological observation networks, especially at the global scale, where traditional grid representations introduce distortions. Graph neural networks (GNNs) address this issue by representing meteorological fields as graphs, where nodes correspond to stations or grid points and edges are used to model spatial dependencies. GNN-based methods have been successfully applied to global and regional forecasting tasks; for example, Keisler [13] used GNNs for six-hour interval forecasting, and GraphCast [15] demonstrated impressive accuracy in global forecasting using a GNN framework. Despite these advances, most existing GNN-based models rely on pairwise spatial relationships, which limits their ability to capture higher-order dependencies inherent in atmospheric processes. To overcome this issue, recent studies have explored more adaptive graph constructions. For example, DGFormer [29] employs a dynamic graph learning mechanism to construct an adjacency matrix via node embedding similarities and optimize the graph structure via physically constrained topological optimization. This dynamic approach allows the inter-station topology to evolve over time, which enhances temporal adaptability and thus improves the performance of long-term forecasts.

In summary, even if the graph structure is adjusted through dynamic mechanisms, the traditional pairwise relationship modeling framework may still be difficult to fully characterize the complex multi-station spatial correlations that are prevalent in meteorological systems. This problem is particularly prominent in short-term and high-impact weather events. Therefore, the limitations of the current method highlight the urgent need for a more structurally expressive modeling framework to more effectively construct and propagate high-order spatial dependencies between meteorological variables, providing basic support for improving the accuracy of short-term forecasts.

2.2. Hypergraph neural networks

To address these limitations, hypergraph neural networks (HGNNs) have emerged as a promising extension of traditional graph-based methods. Unlike conventional graphs, which model pairwise relationships, hypergraphs allow for the representation of higher-order interactions involving multiple nodes, providing a more expressive framework for capturing complex relationships. Early studies focused on adapting GNNs to hypergraph structures by defining hypergraph Laplacian operators and constructing corresponding hypergraph convolution processes. Feng et al. [19] further advanced this methodology by overcoming the limitation of binary relationships in traditional GNNs. Their model integrates higher-order, multi-node associations into the feature extraction process through spectral domain convolution on hypergraphs.

To improve the flexibility of hypergraph models, Gao et al. [30] proposed a generalized hypergraph neural network framework, HGNN+, which improves the representation of complex data relationships. This is achieved through higher-order association modeling and an adaptive hyperedge fusion strategy, improving the processing of multimodal and multi-type data. The two-stage hypergraph convolution mechanism used in HGNN+ markedly enhances the flexibility and representation capacity of information propagation in hypergraphs. Bai et al. [31] introduced a hypergraph attention mechanism that assigns different attention weights to hyperedges during the aggregation process. This mechanism effectively distinguishes the importance of each hyperedge, improving the ability of the model to capture nuanced relationships. Meanwhile, Ding et al. [32] introduced HyperGAT, which combines the attention mechanism with hypergraphs to evaluate the importance of different features during the aggregation process. Zhang et al. [33] presented Hyper-SAGNN, which adapts to heterogeneous hyperedges of arbitrary size through self-attention. The AllSet framework [34] incorporates Deep Sets and Set Transformers into hypergraph neural networks, introducing an innovative approach using multi-set function combinations. Moreover, Wang et al. [35] proposed the T-HyperGNN framework, which models higher-order interactions in hypergraphs by introducing high-dimensional tensor representations. By combining T-spectral and T-spatial convolutions, T-HyperGNN constructs a unified multilevel information transfer mechanism, effectively preserving complex multi-relationships between nodes. Additionally, Ma et al. [36] introduced a multi-view time series hypergraph neural network, which captures higher-order spatial and temporal semantic relationships in human action recognition.

Hypergraph neural networks have shown great promise in a variety of fields, particularly in addressing complex scenarios involving multimodal data and time series forecasting. Building on these advancements, this paper introduces the hypergraph structure to model complex relationships in multivariate time series data.

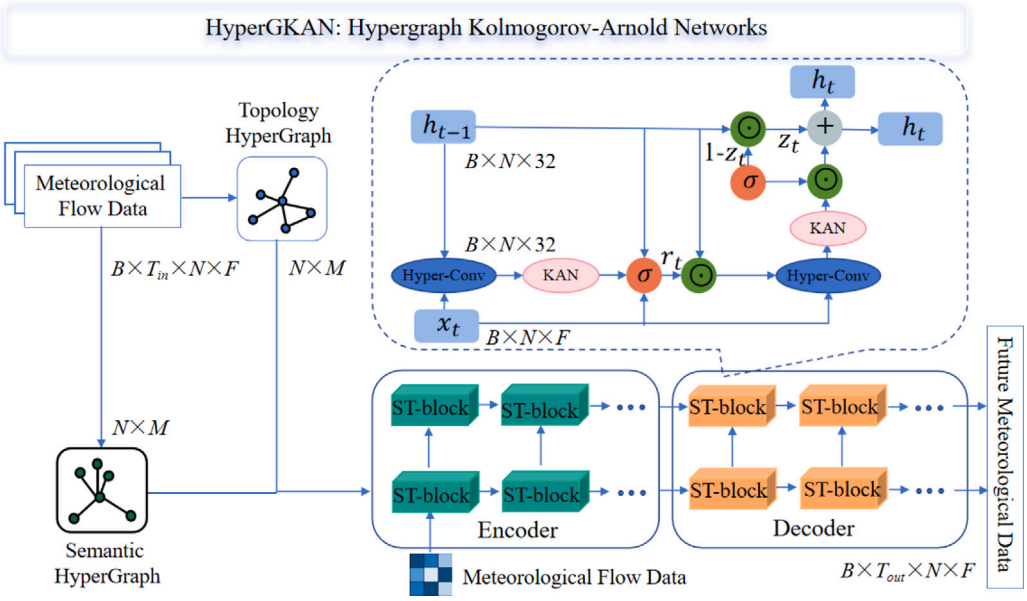

Fig. 1. An overview of the proposed method's framework. The method utilizes the location of weather stations and historical weather data to construct neighborhood hypergraphs and Kolmogorov–Arnold networks. The core of the workflow is an encoder–decoder neural network structure. A spatio-temporal convolution module and hypergraph convolution work with the weather data to process these hypergraphs. The outputs are then merged and presented as detailed forecasts at the station level.

3. Methodology

To address the challenge of multi-step weather prediction and achieve end-to-end forecasting, this paper design a spatio-temporal hypergraph neural network, HyperGKAN,. The model is based on a sequence-to-sequence architecture, where the encoder module extracts key features from historical meteorological time series, and the decoder module uses these features to generate future weather forecasts step-by-step, enabling step-by-step prediction over multiple time steps.

3.1. Problem definition and network architecture

Weather forecasting involves predicting future meteorological conditions by analyzing historical data collected from weather stations. In this paper, the inputs to the HyperGKAN model include the geographical locations of the weather stations and their historical weather data, the data from all weather stations P at the tth time step denoted as, $\{X_t = \{x_1^t, x_2^t, \dots, x_P^t\} \in R^{P\times Q}\}_{t=1}^{T_{in}}$, where T_{in} represents the time dimension and Q is the number of meteorological features. The task of the model is to predict the future time series $\{X_{t+1}, \dots\dots, X_{t+m-1}, X_{t+m}\}$ for m time steps based on the historical time series for k time steps as input $\{X_{t-k}, \dots\dots, X_{t-1}, X_t\}$. Fig. 1 shows the overall framework of HyperGKAN, which is mainly composed of the following three parts:

Input of HyperGKAN: Inputs to the model include geographical longitude and latitude information from N weather stations and historical observation data of consecutive Tin intervals. The input data of each time step has dimensions $B \times N \times F$, where B represents the batch size and F is the number of feature channels for each node. First, the hypergraph structure is constructed from the multivariate time series data by the hypergraph generation module. Using the location information of the weather station, the neighborhood hypergraph is generated by the K-Nearest Neighbor (KNN) method based on the distance metric. Meanwhile, the semantic hypergraph is constructed by the KNN method with semantic metric based on historical data.

Feature extraction: To extract features from meteorological data, historical meteorological data and two $N \times M$ incidence matrices with M hyperedges are input into the encoder–decoder architecture. Each spatio-temporal module combines the modeling advantages of KAN and hypergraph convolutional neural network, and can decouple and combine high-dimensional inputs into one-dimensional functions, thereby reducing the degree of coupling between features and improving the accuracy and expression of feature extraction. After the KAN layer, the model first expands the number of channels to 32 to enhance the feature representation capability, and then reduces it to 16 to further capture the potential relationship between variables.

Encoder–decoder architecture: To model temporal dependencies, GRUs (Gated Recurrent Units) are employed, with the linear layer replaced by HyperGKAN within the GRUs. Building on this design, an encoder–decoder architecture is constructed using multiple GRUs. The encoder's number of layers matches the input time steps, while the decoder's layers correspond to the output time steps. Finally, after the linear layer, an output with the dimension of $B \times T_{out} \times N \times F$ is obtained.

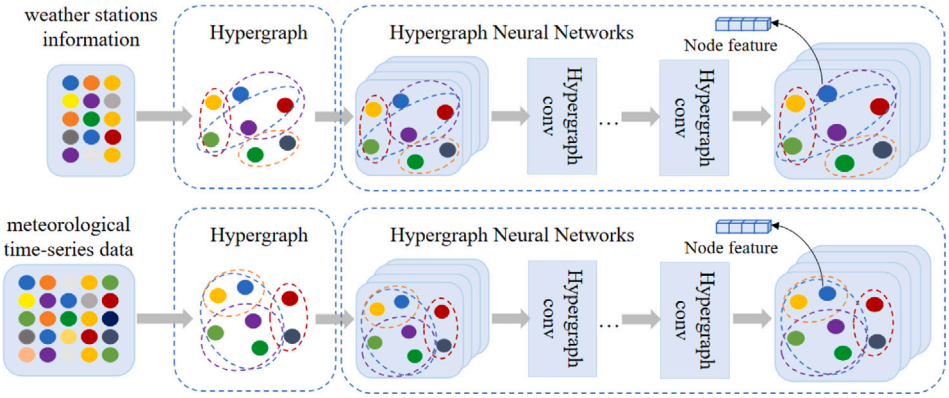

Fig. 2. An architectural flow based on Hypergraph Neural Networks for processing weather station information and meteorological time series data. The whole is divided into two parts, corresponding to two input data types: weather station information and meteorological time series data.

3.2. Adjacency and semantic hypergraph construction

Hypergraphs generalize traditional graphs by allowing each hyperedge to connect more than two nodes, making them ideal for modeling complex relationships between multiple entities. In meteorological data analysis, hypergraphs are particularly useful for capturing both spatial and temporal dependencies between weather stations, which may involve intricate relationships across multiple stations. Each node in the hypergraph represents a weather station, and the hyperedges model interactions between stations based on various factors, such as spatial proximity, semantic similarity, or other meteorological characteristics.

A hypergraph is composed of multiple hyperedges, each containing multiple nodes. The degree of each vertex $v \in \mathcal{V}$ can be expressed as:

$$d(v)_{hg} = \sum_{e \in \mathcal{E}} w(e)\,\mathbf{H}(v,e), \tag{1}$$

where $w(e)$ represent the wight of hyperedge e, $h(v,e)$ denotes whether the hyperedge e contains a node v. For each hyperedge, the hyperedge degree is defined by:

$$\delta(e)_{hg} = \sum_{v \in \mathcal{V}} \mathbf{H}(v,e). \tag{2}$$

In this study, the vertices of the hypergraph, denoted by V, represent weather stations, and the meteorological data from these stations are presented as vertex attributes.

As shown in Fig. 2, multiple hypergraphs have been constructed to uncover the spatial correlations embedded within meteorological data, enabling the capture of multi-spatial information. The construction of these hypergraphs involves two distinct processes: building the adjacency hypergraph G_G and developing the semantic hypergraphs G_S.

Adjacency hypergraph construction. The adjacency hypergraph is constructed to reveal the spatial relationships among nearby weather stations. This structure highlights the presence of effective influence between neighboring stations, attributed to advection and local pattern characteristics of meteorological data. In addition, using the values of meteorological data to calculate dynamic attention for each adjacency hyperedge, which helps mitigate the impact of neighborhood noise in some cases. First, considering that weather stations are located on the Earth's surface, meteorological information from these stations is treated as signals distributed across the surface of a sphere. Each weather station is represented as $\{\mathcal{V} = (v_1, v_2, v_3, \ldots, v_n) : \|\mathcal{V}\| = 1\}$. Then, we calculate the geodesic distance between each pair of weather stations using their latitude and longitude. This distance represents the shortest path between them on the Earth's surface. Utilizing the geodesic distances, the KNN algorithm is employed to identify the nearest $\mathcal{K}$ stations for each weather station and construct adjacency hyperedges $\mathcal{E}_a = (e_j)_{a \in \mathbf{A}}$, $\mathbf{A}$, where $\mathbf{A}$ denotes the number of adjacency hyperedges. The $\mathcal{K}$ closest weather stations to v_i are connected via these hyperedges. Yet, it is important to note that adjacent weather stations might sometimes experience vastly different weather conditions, such as local weather phenomena and severe convective weather events. To effectively aggregate node information, hyperedge weights are computed based on the variability of meteorological features. These weights represent the interaction strength between nodes within each hyperedge. The formula used to calculate hyperedge weights is as follows:

$$w_{ij} = \left(x_i^t - x_j^t\right) / \sum_{m}^{K} \left(x_i^t - x_m^t\right) \quad i, m \in v, \tag{3}$$

where w_{ij} denotes the weight between weather station i and j.

Semantic Hypergraph Construction The adjacency hypergraph represents the spatial correlations within the neighborhood of weather stations based on geographic location. However, weather stations located in distant areas may still exhibit similar

meteorological patterns due to comparable topographic and landform characteristics. This phenomenon is referred to as "semantic correlation". To capture this richer spatial information, the semantic hypergraph is constructed. The semantic hypergraph is analyzed from two perspectives: aggregation and estrangement. The aggregation semantic hypergraph highlights positive and beneficial interactions, signifying useful connections. On the other hand, the estrangement semantic hypergraph indicates dissimilar and potentially disruptive relationships, helping to identify and isolate noisy information.

The variation in topographic can influence the direction of air circulation. Although topographic information generally remains constant over time, it significantly impacts weather patterns. Consequently, historical meteorological data from weather stations can serve as an indirect representation of topographic information. Specifically, the hypergraph is constructed using the Euclidean distance (ED) between the meteorological time-series data of the stations. For each station, the most similar $\mathcal{K}$ stations are selected to form semantic topographic hyperedges $\mathcal{E}_t = \left(e_t\right)_{t\in\mathbf{T}}$. Thus, the semantic hypergraph G_S illustrates the "topographic similarity" among weather stations.

In summary, adjacency hypergraph G_G and semantic hypergraph G_S are constructed.

3.3. KAN-based HyperGNN layers

The Kolmogorov–Arnold Network (KAN), rooted in the Kolmogorov–Arnold representation theorem, is a neural network architecture designed to approximate complex nonlinear systems. The theorem states that any continuous multivariate function can be represented using combinations of one-dimensional continuous functions. By leveraging this principle, KAN enables the approximation of multivariate functions using nested transformations of univariate functions.

In this method, the KAN architecture serves as an alternative to the MLP, drawing on the Kolmogorov–Arnold representation theorem, which states that any arbitrary multivariate function $f : [0,1]^d \to \mathbb{R}$ can be represented in the following form:

$$f(x) = \sum_{i=1}^{2d+1} \Phi_i(\sum_{j=1}^{d} \phi_{ij}(x_j)), \tag{4}$$

where all the Φ and ϕ functions are univariate and their combination uniquely constitutes a multivariate operator.

This representation can be understood as a two-step process: first a different univariate nonlinear activation function is applied to each dimension of the input, and then the outputs of these functions are summed. Based on this, the KAN layer maps the space $A \subseteq \mathbb{R}^d$ to the space $B \subseteq \mathbb{R}^{d'}$, similar to the mapping of the MLP layer. The layer consists of $d \times d'$ trainable univariate functions $\{\phi_{ij}, 1 \le i \le d', 1 \le j \le d\}$. For input $x \in A$, its output x' is computed as:

$$x'_i = \sum_{j=1}^{d} \Phi_{ij}((x_j)), \tag{5}$$

The KAN architecture supports the input dimension d, the output dimension d', and the model constructed by the formulae as above is a universal function approximator. Compared with MLP, the KAN architecture has some advantages in terms of complexity. According to the theorem, any multivariate function from $[0,1]^d$ to $\mathbb{R}^{d'}$ requires at most $(2d^2+d)\times d'$ univariate functions, whereas the universal approximation theorem of MLP may require an infinite number of neurons. However, the behavior of these univariate functions can be very complex (e.g., fractal or non-smooth) and may be difficult to represent or learn.

To address this problem, MLP mitigates the need for infinite width by stacking layers of finite width; similarly, KAN reduces the complexity requirement for nonlinearity by stacking multiple KAN layers. Thus, the final output of the function is:

$$f_{out}(x) = KAN(x) = \Phi_L \circ \Phi_{L-1} \circ ... \circ \Phi_1, \tag{6}$$

where, L denotes the number of KAN layers.

KAN achieves superior nonlinear representation by replacing fixed activation functions with learnable univariate spline functions. For a KANLinear layer with input dimension d_{in} and output dimension d_{out}, the time complexity is dominated by spline basis computation and linear transformations. The B-spline basis calculation requires $O(d_{\text{in}} \cdot (G+S))$ operations, where G is the grid size (default $G = 5$) and S is the spline order (default $S = 3$). Subsequent linear operations introduce $O(d_{\text{in}} \cdot d_{\text{out}} \cdot (G+S))$ complexity. While this exceeds the $O(d_{\text{in}} \cdot d_{\text{out}})$ complexity of an MLP layer, KAN's shallow architecture compensates by reducing the need for deep stacking. The space complexity stems from storing $d_{\text{in}} \cdot d_{\text{out}} \cdot (G+S)$ spline coefficients and $d_{\text{in}} \cdot d_{\text{out}}$ base weights. With default parameters ($G = 5$, $S = 3$), a KANLinear layer requires 8× more parameters than an equivalent MLP layer, but fewer layers are needed to achieve comparable nonlinearity.

3.4. KAN-based multi-information HyperGNN

In weather forecasting, the hypergraph convolution kernel enables efficient spatial information transfer by combining features from various meteorological data sources, offering a robust approach to modeling spatial dependencies. Building on this, a novel variant of the hypergraph model is introduced by integrating Kolmogorov–Arnold Networks (KANs) for transforming node features. Instead of using traditional fully-connected layers or multilayer perceptrons (MLPs), KAN layers are utilized to construct a new type of Hypergraph Convolutional Layer, the HyperGKAN layer. This approach allows the model to effectively capture complex spatial dependencies and adaptively transform node features, enhancing its capability to represent meteorological data and improving prediction accuracy.

Algorithm 1 Prediction framework of HyperGKAN

Input: Historical meteorological dataset $\mathbf{X}$, initial traffic network G.
Output: Prediction results $\mathbf{P}$.
Construct neighborhood hypergraph G_G using KNN based on geodesic distances.
Construct semantic hypergraph G_S using KNN based on feature similarity of X.
for *iter* < *iters* **do**
    Dynamically fuse spatial and semantic features using KAN-based hypergraph convolutional layers.

$$y_i \leftarrow f(\mathrm{conv}(G_G, G_S))$$

    Calculate prediction loss L_{MAE} using predicted values P_i and true values P_i as:

$$\mathcal{L}_{\mathrm{MAE}} \leftarrow \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

end for
Output Prediction Results: Generate final prediction results P based on the output of the KAN-enhanced hypergraph network.

The strength of KAN lies in its ability to approximate arbitrary multidimensional continuous functions through a series of nonlinear transformations, based on the Kolmogorov–Arnold representation theorem. By leveraging this powerful feature, the HyperGKAN layer captures the intricate relationships among node features during the transformation process, resulting in a more flexible representation of higher-order correlations in the hypergraph. This is particularly useful when dealing with complex weather data.

The specific implementation of the HyperGKAN layer is as follows:

$$X_g^l = \Phi^l(D_v^{-1/2} H_g W_g D_e^{-1} H_g^\top D_v^{-1/2} X_g^{l-1}), \tag{7}$$

$$X_s^l = \Phi^l(\tilde{D}_v^{-1/2} H_s W_s \tilde{D}_e^{-1} H_s^\top \tilde{D}_v^{-1/2} X_s^{l-1}), \tag{8}$$

where $X_g^{l-1} \in \mathbb{R}^{N\times C}$ and $X_s^{l-1} \in \mathbb{R}^{N\times C}$ represent neighborhood features and semantic features in layer l respectively. N indicates the number of meteorological stations and C denotes the dimension of each weather station feature. Φ represents single KAN layer.

The neighborhood features and semantic features obtained after convolution are dynamically fused through a linear layer. Ultimately, the fusion representation of the multivariate time series can be expressed as:

$$X_O = Linear(Concat(X_G, X_S)), \tag{9}$$

where X_G and X_S denote the post-convolution neighborhood and semantic features, respectively, and *Atten* denotes the operation of the attention mechanism. This fusion step ensures that the model effectively integrates multi-dimensional weather data for improved forecasting accuracy.

In summary, the complete training process of our proposed method is outlined in Algorithm 1.

4. Experiments

All experiments were performed on a system with an Intel® Xeon® Gold 6326 CPU @ 2.90 GHz, an NVIDIA A800 80 GB GPU, and Ubuntu 18.04.1 LTS, with the model implemented in the PyTorch framework. To assess the performance of the HyperGKAN model, we conducted comprehensive experiments on four weather datasets, validated the results through ablation studies and various visual analyses, and examined the impact of the number of nodes per hyperedge.

4.1. Experiment setup

Experiments used the WeatherBench high-resolution weather dataset, comprising 2048 global distributed nodes to record real weather observations across diverse regions and varying resolutions. This study focuses on hourly weather forecasts using the subset shared by Lin et al. [12]. The subset includes global meteorological data from January 1, 2010, to December 31, 2018, covering variables like temperature, cloud cover, humidity, and surface wind speed. For detailed analysis, the forecasts of these four meteorological variables are compared separately. Both the input and prediction time steps for the HyperGKAN model are set to 12 h. The dataset is divided into 2300 training samples, 329 validation samples, and 657 test samples.

HyperGKAN is a spatio-temporal hypergraph convolutional network designed for weather forecasting. To validate the effectiveness of our experimental comparisons, we primarily focused on comparing it with other spatio-temporal graph convolutional networks. The specifics of the comparison methods are described below:

- GCGRU [37] extracts spatial and temporal dependencies step by step at each time step by combining the properties of GCN and GRU.

Table 1
Comparison of our method with mainstream spatio-temporal graph neural networks on four weather datasets. Results in bold indicate the best overall performance.

Method	Venue	Temperature		Cloud cover		Humidity		Wind	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
GCGRU	ICLR_17	1.1757	1.9776	0.1584	0.2547	5.1018	8.0834	1.4102	2.3073
STGCN	IJCAI_18	3.3083	5.7367	0.2019	0.2954	7.7572	8.1960	3.6477	4.8143
DCRNN	ICLR_18	1.1071	1.9391	0.1587	0.2554	5.0717	8.0047	1.3971	2.2758
ASTGCN	AAAI_19	1.4766	2.4599	0.1986	0.2948	6.4717	9.0379	2.0883	3.1344
AGCRN	NIPS_20	1.2471	1.9095	0.1603	0.2589	5.6321	9.1527	2.3045	3.3044
CLCRN	AAAI_22	1.0719	1.7320	0.1495	0.2438	**4.5087**	**7.0046**	1.2362	1.9641
SimpleSTG	TMLR_23	**0.8813**	**1.5434**	0.1561	0.2516	4.7718	7.5138	2.4906	3.7017
FCSTGNN	AAAI_24	1.0326	1.7248	0.1654	0.2635	5.0428	7.9048	1.4326	2.3179
HyperGKAN	–	0.8942	1.5747	**0.1455**	**0.2420**	4.7059	7.4285	**1.0029**	**1.5702**

- STGCN [38] uses one-dimensional convolution along the temporal dimension to capture temporal dependencies, with time-series data as input. This time-series convolution effectively captures both short-term and long-term temporal features within a fixed time window.
- DCRNN [39] is based on the concept of crossover networks, where features are cross-processed through RNN networks. It introduces a partial parameter sharing structure, which effectively captures the underlying correlations between different tasks, improving learning efficiency and optimization for multi-task scenarios.
- ASTGCN [40] and MSTGCN [40] utilize spatial attention mechanisms to learn spatial correlations and temporal attention to capture temporal correlations across different time steps.
- TGCN [41] achieves efficient modeling of spatio-temporal dependencies through type-aware neighbor aggregation, CNN-based multilevel feature interaction updating and TransTag regularization to capture semantic information of user-tag-item triples.
- AGCRN [42] dynamically models the spatial and temporal dependencies of nodes by integrating the Node Adaptive Parameter Learning (NAPL) module and the Data Adaptive Graph Generation (DAGG) module. The NAPL module learns node-specific feature patterns, while the DAGG module generates adaptive adjacency matrices to capture the dynamic spatio-temporal correlations between nodes.
- CLCRN [12] captures the local spatial patterns of each node through a conditional local convolution kernel, which is embedded into a recurrent neural network to model the temporal dynamics, thus capturing the spatial and temporal features of meteorological data in a spatio-temporal dependency.
- SimpleSTG [43] captures spatio-temporal dependencies by combining spatial and temporal modules into a unified framework, incorporating a simplified Neural Architecture Search (NAS) strategy based on stochastic search.
- FCSTGNN [44] enables comprehensive spatio-temporal dependency modeling of multi-sensor time series data by constructing FC graphs to establish complete connectivity between all sensors and timestamps, and by convolving moving pooled GNN layers with sliding windows.

In the experiments, The model was trained using the Adam optimizer with an initial learning rate of 0.01, which followed a step-decay schedule at predefined intervals. Both training and validation phases utilized a batch size of 16. The loss function was defined as Mean Absolute Error (MAE), and the activation function in the Kolmogorov–Arnold Network (KAN) layers was set to SiLU (Sigmoid Linear Unit). No explicit L1 or L2 regularization was applied during training. The total number of training epochs was capped at 500, with early stopping triggered if no improvement in validation loss was observed for 35 consecutive epochs. The model's effectiveness was evaluated using two metrics: mean absolute error (MAE) and root mean square error (RMSE), as shown below:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|\left(y_i^t - \hat{y}_i^t\right)\right| \tag{10}$$

$$RMSE = \frac{1}{N}\left(\sum_{i=1}^{N}\left(y_i^t - \hat{y}_i^t\right)^2\right)^{1/2} \tag{11}$$

Here, y_i^t represents the predicted value at time t for the ith meteorological observation station, and $\hat{y}_i^t$ is the corresponding ground truth value. The MAE measures the average absolute errors between predicted values and the ground truth, reflecting their magnitude of difference. The $RMSE$ calculates the square root of the mean squared differences, providing a measure of the model's stability.

4.2. Performance comparison

Table 1 demonstrates the comparison results of our proposed HyperGKAN method with other mainstream spatio-temporal graph neural networks on four different meteorological datasets. The specific evaluation metrics include the Mean Absolute Error (MAE)

Table 2
Experimental comparison of this method with the CLCRN method for different prediction time steps.

Method	Temperature		Cloud cover		Humidity		Wind	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
CLCRN(3 h)	0.3910	0.7116	0.0925	0.1619	**1.9726**	**3.3682**	0.4975	0.8603
Our(3 h)	**0.3654**	**0.7032**	**0.0902**	**0.1540**	2.0906	3.6335	**0.4954**	**0.8580**
CLCRN(6 h)	0.6673	1.1771	0.1202	0.2025	**3.1267**	**5.1698**	0.7966	1.3288
Our(6 h)	**0.6599**	**1.1332**	**0.1114**	**0.1904**	3.6916	6.0886	**0.7896**	**1.3072**
CLCRN(12 h)	1.0719	1.7320	0.1495	0.2438	**4.5087**	**7.0046**	1.2362	1.9641
Our(12 h)	**0.8942**	**1.5747**	**0.1455**	**0.2420**	4.7059	7.4285	**1.0029**	**1.5702**

and Root Mean Square Error (RMSE) for Temperature, Cloud cover, Humidity, and Wind. Smaller values of MAE and RMSE indicate lower prediction errors and better model performance.

As shown in Table 1, HyperGKAN delivers strong predictive performance on cloud cover and wind speed, outperforming all baselines. However, it performs slightly below SimpleSTG on the temperature dataset. This can be attributed to HyperGKAN's architectural emphasis on capturing global and high-order spatial relationships, whereas SimpleSTG is more adept at modeling localized spatial patterns. In the case of humidity, where local correlations between adjacent meteorological sites dominate, HyperGKAN's use of non-local dependencies may introduce redundant information, resulting in marginally reduced effectiveness.

These performance variations align with the inherent spatial dynamics of different meteorological variables. For instance, wind and cloud cover are often influenced by large-scale atmospheric systems and synoptic processes that span extensive geographical areas. In such contexts, HyperGKAN's hypergraph structure, particularly its ability to connect semantically similar but geographically distant nodes via hyperedges, proves advantageous by capturing high-order spatial correlations beyond pairwise interactions.

In contrast, temperature and humidity are governed by more localized phenomena – such as topography, microclimates, and surface-atmosphere exchanges – where short-range spatial dependencies are dominant. While HyperGKAN incorporates local neighborhood hyperedges, the inclusion of global semantic hyperedges may dilute the predictive power for variables with concentrated spatial dependencies. Traditional graph models, which focus on pairwise proximity, may be more aligned with the spatial behavior of these variables.

Nevertheless, HyperGKAN consistently ranks among the top-performing models across all tasks, securing either the first or second position in each category (as shown in Table 1). Its balanced integration of local and global spatial modeling enables effective handling of diverse meteorological patterns. Rather than exposing a limitation in generalizability, the observed performance variations underscore the model's adaptability to different spatial correlation structures inherent in atmospheric data.

Table 2 demonstrates the comparison of the results at different time steps. It can be seen that our proposed method outperforms CLCRN in the assessment of most meteorological elements. However, the performance of humidity prediction is weaker, mainly due to the fact that humidity is highly dependent on multiple factors, such as temperature, barometric pressure, precipitation, etc., and its variations are subject to time-delayed effects and significant localized variations, which increase the difficulty of the model prediction. Overall, our method effectively handles spatial information at each time step, which significantly improves the accuracy of the long-term prediction.

4.3. Ablation analysis

We conducted ablation tests to validate the effectiveness of the HyperGKAN design. Five ablation variants were evaluated on temperature, cloud cover, humidity, and wind datasets. The implemented model variants are described as follows:

(1) w/o-G: Replaces the hypergraph convolution layer with a conventional graph convolution layer.

(2) w/o-Nei: Removes the neighborhood hypergraph, relying solely on the semantic hypergraph to model inter-sequence dependencies.

(3) w/o-Sem: Removes the semantic hypergraph, relying solely on the neighborhood hypergraph to model inter-sequence dependencies.

(4) w/o-KAN: Replaces the KAN with MLP for learning feature representations.

(5) w/o-Fusion: Replaces the attention fusion module with a simple addition operation for feature fusion.

Table 3 demonstrates a comparison of the performance of the different variants of the model on the temperature, cloud cover, humidity and wind datasets. The results of the analyses are as follows:

(1) The results of the w/o-G variant show a significant increase in MAE and RMSE in all four tasks. This indicates:

- The hypergraph convolutional layer is able to capture higher-order node relationships, whereas the traditional graph convolution can only model first-order neighborhood dependencies.
- For meteorological data, complex diffuse relationships often exist between neighboring sites, and these higher-order relationships cannot be captured by traditional graph convolution.

(2) The results of the w/o-Nei variant show a certain degree of performance degradation on the four prediction tasks of the meteorological factors. This indicates:

Table 3
Experiments with different forecasting time steps are compared with the latest CLCRN method.

Method	Temperature		Cloud cover		Humidity		Wind	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
w/o-G	0.9331	1.6360	0.1565	0.2573	4.9931	7.8788	1.5255	2.3876
w/o-Nei	1.0293	1.8047	0.1625	0.2631	5.1798	8.1934	1.3002	1.9141
w/o-Sem	1.0315	1.8369	0.1558	0.2562	5.3106	8.2622	1.1189	1.6337
w/o-KAN	1.0174	1.6688	0.1465	0.2458	4.5067	7.0519	1.2201	1.8922
w/o-Fusion	0.9310	1.6515	0.1561	0.2812	4.9337	7.7684	1.1018	1.6032
Our	**0.8942**	**1.5747**	**0.1455**	**0.2420**	**4.7059**	**7.4285**	**1.0029**	**1.5702**

- Neighborhood hypergraphs are able to capture local time series dependencies, such as weather propagation patterns between neighboring weather stations.
- Removing the neighborhood hypergraph weakens the model's ability to model short-time local relationships, leading to an increase in prediction error.

(3) The results of the w/o-Sem variant indicate that semantic hypergraphs positively impact elemental prediction. This shows:

- Semantic hypergraphs are used to capture global semantic dependencies, such as long-distance correlations between weather stations with similar weather patterns.
- Removing the semantic hypergraph makes it difficult for the model to effectively capture these global features, which affects the prediction performance.

(4) The results of the w/o-KAN variant show that this variant performs poorly on all four data elements. This indicates:

- The KAN layer is capable of modeling complex univariate features using learnable non-linear functions, which plays an important role in spatio-temporal feature extraction for meteorological data.
- The inability of the linear layer to capture such non-linear features leads to a reduction in the model's ability to model complex spatio-temporal dynamics.

(5) The results of the w/o-Fusion variant show that replacing the feature attention fusion module with a simple addition operation leads to a decrease in model performance. This indicates:

- The attention fusion module is able to dynamically adjust the contribution of each feature by assigning weights according to the importance of different features.
- Simple addition operations may lead to information conflict or redundancy and fail to fully utilize the synergistic relationship between features.

The ablation experiment verifies the importance of each module in the whole framework. The removal of any module leads to different degrees of degradation in prediction performance, indicating that the designed framework has a clear functional division of labor and synergy among the various parts, which is a key factor in achieving efficient weather prediction.

Given that the traditional multi-layer perceptron (MLP) not only has linear modeling capabilities, but also has significantly enhanced nonlinear expression capabilities after the introduction of specific nonlinear activation functions (such as sine function or sigmoid function), especially in dealing with complex spatiotemporal dependencies and nonlinear mapping tasks. In order to comprehensively evaluate the effectiveness and superiority of the proposed Kolmogorov–Arnold Network (KAN) in modeling nonlinear features, we designed a set of MLP structures based on different activation functions as comparison models and conducted systematic ablation experiments. The specific settings are as follows:

(1) w-ReLU: ReLU is used as the activation function in MLP to improve the sparsity and nonlinear modeling capabilities of the network for feature representation;

(2) w-Sine: The Sine activation function is used to capture the periodic fluctuation characteristics in the data, which is suitable for modeling physical processes with oscillatory behavior;

(3) w-Sigmoid: The Sigmoid activation function is used to introduce smooth nonlinear transformations, so that the output is bounded within a fixed interval, which helps stabilize the training process and limit the impact of outliers.

From the experimental results in Table 4, we can see that although MLP has certain expressive power after using nonlinear activation functions such as Sine, ReLU or Sigmoid, it has not surpassed the KAN-based model in all tasks, especially in the wind speed and cloud cover prediction tasks, where KAN shows obvious advantages. This shows that the nonlinear ability of traditional MLP is still limited by the form of fixed activation functions and the deep stacking of network structures. KAN can more flexibly approximate complex high-dimensional nonlinear functions by introducing learnable one-dimensional function combinations, and has stronger expressive power in extracting spatiotemporal dynamic features and capturing local and global dependencies. In addition, we further verified the importance of the KAN module in the ablation experiment in Section 4.3. After removing KAN, the MAE and RMSE of the model in all prediction tasks increased significantly, indicating that it plays a key role in modeling complex meteorological data. In theory, KAN is based on the Kolmogorov–Arnold representation theorem and decomposes multivariate

Table 4
Ablation experiment on structural of multilayer perceptron with different activation functions.

Method	Temperature		Cloud cover		Humidity		Wind	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
w-Sine	0.9347	1.6499	0.1547	0.2564	5.0359	7.9023	1.2737	2.0955
w-ReLU	0.9194	1.6330	0.1548	0.2563	5.0199	7.8785	1.2719	2.0882
w-Sigmoid	1.17006	1.7249	0.1547	0.2568	4.9851	7.8258	1.2722	2.0913
Our	**0.8942**	**1.5747**	**0.1455**	**0.2420**	**4.7059**	**7.4285**	**1.0029**	**1.5702**

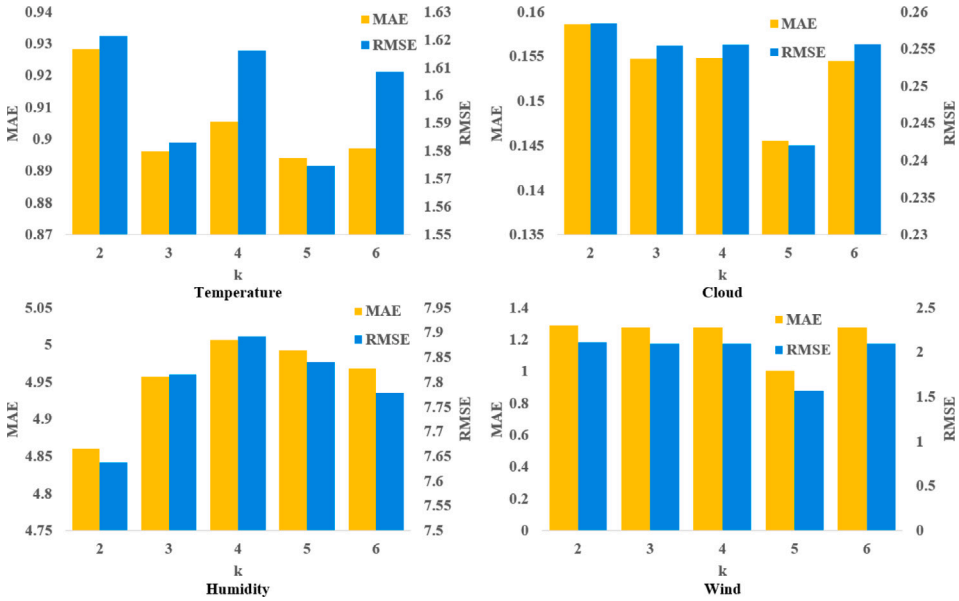

Fig. 3. Experiments with different K of hyperedges in meteorological variables datasets.

functions into nested one-dimensional function combinations. This structured nonlinear modeling method has both strong expressive power and structural interpretability. Therefore, combining theoretical analysis with empirical results, it can be fully demonstrated that KAN has stronger fitting and generalization capabilities than traditional MLP, and is an effective means to achieve high-precision weather forecasting.

4.4. Effect of hyperedges on prediction performance

The number K of nodes within each hyperedge is a key parameter for constructing K-nearest neighbor hypergraphs, smaller K values mean more localized pairwise relationships, while larger K values enable the model to capture higher-order dependencies between nodes. We conducted K-sensitivity experiments with K values ranging from 2 to 6, evaluating the impact on prediction performance for temperature, cloud cover, humidity, and wind, as summarized in Fig. 3.

When $K = 2$, the hypergraph structure essentially degenerates into a standard graph structure, and the model performance is similar to that of traditional GNN, which makes it difficult to capture high-order semantic information. As K increases, the prediction accuracy of the model on multiple data sets is significantly improved. Specifically, when $K = 5$, the model achieves the best MAE and RMSE performance in temperature, cloud cover, and wind prediction tasks, indicating that moderately sized hyperedges are more helpful in modeling the complex spatiotemporal dependencies in these variables. However, when K further increases to 6, the model performance decreases to a certain extent, which may be due to the introduction of noise information by excessively large hyperedges.

It is worth noting that humidity prediction performs best when $K = 2$, which is closely related to the strong local differences and rapid changes in humidity between different geographical locations. This result shows that a relatively smaller hyperedge scale is more effective in modeling meteorological variables with strong local dependencies.

In summary, the experimental results verify the importance of introducing appropriate high-order structures to improve the prediction ability of meteorological elements, and further prove the effectiveness of our proposed hypergraph construction strategy in balancing structural locality and expression ability. This also highlights the advantages of HyperGKAN over traditional GNN methods in modeling complex spatiotemporal relationships.

5. Discussion

While the proposed HyperGKAN model demonstrates competitive performance across multiple tasks, there remain opportunities to enhance its effectiveness. It is worth noting that the current dataset used in this study, while widely adopted within the research community, is relatively limited in both scale and spatial resolution. This constraint may hinder the full expression of the hypergraph structure's advantages, especially in modeling high-order dependencies and long-range semantic correlations. It is expected that applying HyperGKAN to larger-scale meteorological datasets with richer spatial coverage would better exploit the strengths of hypergraph modeling and yield further performance improvements.

6. Conclusion

This paper proposes a Hypergraph Kolmogorov–Arnold Networks (HyperGKAN) for station level spatio-temporal weather prediction. The model combines neighborhood hypergraph and semantic hypergraph structures, which can effectively model high-order spatial dependencies in meteorological data and enhance the ability to describe complex dynamic processes such as atmospheric advection and diffusion. At the same time, the introduction of KAN's learnable one-dimensional function replaces the fixed weights in traditional graph convolution, making the feature propagation process more flexible, thereby improving the model's spatiotemporal modeling ability and expressiveness. Experiments on four types of meteorological element data sets have shown that HyperGKAN outperforms existing methods in cloud cover and wind speed prediction tasks, and has also achieved competitive results in temperature and humidity prediction, showing higher overall prediction accuracy.

It should be pointed out that compared with medium- and long-term forecasting models such as GraphCast, NeuralGCM, and Pangu-Weather that rely on large-scale reanalysis data such as ERA5, this work is limited by the data time span and meteorological variable dimensions, and mainly focuses on short-term forecasting tasks within 12 h. In the future, we will consider introducing a larger-scale, multivariate reanalysis dataset to expand the model's application potential in medium- and long-term weather forecasting, and further introduce more business-significant indicators such as the anomaly correlation coefficient (ACC) to comprehensively evaluate the model's performance at different spatio-temporal scales.

In addition, although HyperGKAN has achieved significant improvements in accuracy, its hypergraph construction and semantic similarity calculation introduce additional computational burdens, and there is a certain amount of overhead in both the training and reasoning stages. The current version of the model takes accuracy optimization as its main goal, and has not yet been specifically designed for efficiency. Subsequent research will focus on building a more efficient model structure or introducing an approximate reasoning mechanism to improve operational efficiency while maintaining forecasting performance.

CRediT authorship contribution statement

Jian Tang: Writing – review & editing, Writing – original draft. **Kai Ma:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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